H4 H4_timeseries environmental-variable selection

STEP 1
Inventory (231)

UCM			Atmo raw	Atmo average	LYS1 raw	LYS2 raw	LYS average	GVI imagery	SVF imagery	Sound			Solar	Empatica		Eye-tracker	Questionnaire	Metadata / QC			
AIR RH	GPS speed kmh	OPT	left pm1	pm1 avg	LYS1 kelvin	LYS2 kelvin	kelvin avg	GVI 50m	SVF	band proportion 20-200 Hz	octave proportion 62.5 Hz	vehicle score	activeness	acc x	hrv td rmssd	blink duration s	current comfort 1 7	age	Date (Atmo/LYS)	height m	PhaseID
AIR T mid	HUMIDEX SR	PET SR	left pm10	pm10 avg	LYS1 lux	LYS2 lux	lux avg	Grass phase mean	Building %	band proportion 200-2000 Hz	octave proportion 8000 Hz	wind proportion	BVF	acc y	hrv td sdnn	distance from center	environmental comfort mean 1 5	bike frequency	Date (eye)	HRV 5min available	regular physical activity
HUMIDEX	IO flag	SUN alt	left pm2.5	pm2.5 avg	LYS1 lux cat	LYS2 lux cat	medi avg	GVI grass 50m	Ceiling %	band proportion 2000-20000 Hz	rural natural proportion	wind score	Clearness Index	acc z	met	eyelid aperture avg	pleasantness 1 5	bmi category who	Date (UCM)	HRV 5min max gap (s)	residence type
MRT	IR back	SUN az	right pm1		LYS1 b'	LYS2 b'		GVI image capture date	Grass %	category	rural natural score		Clearness Index G0	eda phasic	skin temperature	fixation duration s	prs11 being away mean 1 5	bmi kg m2	Datetime	HRV 5min removed pct	residential area
PET	IR down	UTCI SR	right pm10		LYS1 g'	LYS2 g'		GVI phase mean	Plant %	dominant frequency (Hz)	spectral centroid (Hz)		Clearness Index Infer	eda scl usiemens	vector magnitude	fixation rate	prs11 coherence mean 1 5	clothing accessories	end time s	HRV 5min segment end	response index
SND dBA	IR front	WIND AWA	right pm2.5		LYS1 medi	LYS2 medi		GVI plant 50m	Sky %	mechanisms proportion	spectral rolloff 85% (Hz)		fitness	eda tonic		gaze centrality	prs11 fascination mean 1 5	clothing footwear	gender	HRV 5min segment start	segment
SUN Gh	IR left	WIND TWD	left humidity		LYS1 movement	LYS2 movement		GVI tree 50m	Total Builtup %	mechanisms score	unidentify proportion		TVF	heart rate		gaze dispersion	prs11 scope mean 1 5	clothing height raw	GPS alt	HRV 5min valid rri count	site familiarity 1 10
UTCI	IR right	WIND TWS	left temperature		LYS1 r'	LYS2 r'		Plant phase mean	Total Vegetation %	octave proportion 1000 Hz	urban manmade negative proportion			hrv fd hf		gaze velocity	prs11 total mean 1 5	clothing lower body	GPS hdg	HRV 5min valid span (s)	SVF Frames In Window
WIND AWS	IR spot gnd		right humidity		LYS1 rgbb	LYS2 rgbb		Tree phase mean	Tree %	octave proportion 125 Hz	urban manmade negative score			hrv fd hfnu		pupil change rate	space familiarity 1 5	clothing skin tone	GPS HDOP	MAG hdg	SVF Status
AH	IR spot left		right temperature		LYS1 rgbg	LYS2 rgbg			Wall %	octave proportion 2000 Hz	urban manmade neutral proportion			hrv fd lf		pupil diameter avg	stai6 mean 1 5	clothing upper body	GPS lat	nationality	vision correction
AIR T bot	IR spot right				LYS1 rgbir	LYS2 rgbir				octave proportion 250 Hz	urban manmade neutral score			hrv fd lf hf ratio		saccade amplitude	thermal comfort 1 5	clothing weight raw	GPS lon	own bike for experiment	weight kg
AIR T top	IR spot sky				LYS1 rgbr	LYS2 rgbr				octave proportion 31.25 Hz	urban manmade positive proportion			hrv fd lfnu		saccade duration s		commute modes	GPS speed valid sample count	participant id	years in brussels
AIR temp	IR up									octave proportion 4000 Hz	urban manmade positive score			hrv fd total power		saccade peak velocity		daily walking	GVI match dist m	ParticipantID	
fp	MRT S									octave proportion 500 Hz	vehicle proportion			hrv fd vlf		stress composite		Date	has children	pct complete	

Parameters and decisions by source

1. UCM
- Retained: AIR RH, AIR T mid, HUMIDEX, MRT, PET, SND dBA, SUN Gh, UTCI, WIND AWS.
- AH: Humidity representative: ucm_AIR_RH; AH is an alternative formulation
- AIR T bot, AIR T top, AIR temp: Air-temperature representative: ucm_AIR_T_mid; alternate height/aggregate channels not co-entered
- fp: The definition, units and intended interpretation of fp have not been verified against source documentation; retain in the inventory but exclude from Step 2 analytical eligibility.
- GPS speed kmh: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.
- HUMIDEX SR, MRT S, PET SR, UTCI SR: Derived alternate UCM formulations (_SR/_S suffix); use the corresponding unsuffixed UCM measure
- IO flag: Discrete indoor/outdoor setting flag excluded; experimental phase protocol already stratifies outdoor travel routes.
- IR back, IR down, IR front, IR left, IR right, IR spot gnd, IR spot left, IR spot right, IR spot sky, IR up: Directional IR radiometer channels (inputs used to compute ucm_MRT [CityFeel17]); measured solar exposure representative is ucm_SUN_Gh; not co-entered as separate radiation predictors
- OPT: Derived/redundant: ucm_OPT is operative temperature, derived from air temperature and MRT [CityFeel17]
- SUN alt, SUN az: Declared Step 2 exclusion: solar altitude and solar azimuth are outside the selected exposure set
- WIND AWA, WIND TWD: Derived wind direction/angle from ucm_WIND_AWS; wind speed representative is ucm_WIND_AWS. Circular direction coding is outside the current linear wind-speed specification.
- WIND TWS: Derived true-wind speed (device correction from AWS and related fields); wind representative: ucm_WIND_AWS

2. Atmo raw
- Inputs to retained averages: left pm1, left pm10, left pm2.5, right pm1, right pm10, right pm2.5. Highlighted as source measurements; not entered alongside their averages.
- left humidity, right humidity: Raw-source inventory only: UCM ucm_AIR_RH is the declared relative-humidity measure
- left temperature, right temperature: Raw-source inventory only: UCM ucm_AIR_T_mid is the declared ambient-temperature measure

3. Atmo average
- Retained: pm1 avg, pm10 avg, pm2.5 avg.

4. LYS1 raw
- Inputs to retained averages: LYS1 kelvin, LYS1 lux. Highlighted as source measurements; not entered alongside their averages.
- LYS1 lux cat: Derived illuminance category; continuous lux is available; no averaged replacement exists.
- LYS1 b', LYS1 g', LYS1 r', LYS1 rgbb, LYS1 rgbg, LYS1 rgbir, LYS1 rgbr: Exported RGB/IR or r'/g'/b' channel; no average or conversion to an independently calibrated H4 exposure is declared. Kept in the inventory, not falsely claimed to be represented by lux. Validate units/calibration and a spectral hypothesis before modeling.
- LYS1 medi: Source of lys_medi_avg; not co-entered with its own deterministic average. This is a study aggregation choice, not evidence that placement bias was eliminated. LYS1/LYS2 measure different wearer locations.
- LYS1 movement: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.

5. LYS2 raw
- Inputs to retained averages: LYS2 kelvin, LYS2 lux. Highlighted as source measurements; not entered alongside their averages.
- LYS2 lux cat: Derived illuminance category; continuous lux is available; no averaged replacement exists.
- LYS2 b', LYS2 g', LYS2 r', LYS2 rgbb, LYS2 rgbg, LYS2 rgbir, LYS2 rgbr: Exported RGB/IR or r'/g'/b' channel; no average or conversion to an independently calibrated H4 exposure is declared. Kept in the inventory, not falsely claimed to be represented by lux. Validate units/calibration and a spectral hypothesis before modeling.
- LYS2 medi: Source of lys_medi_avg; not co-entered with its own deterministic average. This is a study aggregation choice, not evidence that placement bias was eliminated. LYS1/LYS2 measure different wearer locations.
- LYS2 movement: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.

6. LYS average
- Retained: kelvin avg, lux avg.
- medi avg: Derived metric estimated from LYS RGB channels, not measured spectrally; retained light measures lys_lux_avg and lys_kelvin_avg

7. GVI imagery
- Retained: GVI 50m.
- Grass phase mean, GVI grass 50m, GVI image capture date, GVI phase mean, GVI plant 50m, GVI tree 50m, Plant phase mean, Tree phase mean: Vegetation representative: gvi_50m; alternative scales/components not co-entered

8. SVF imagery
- Retained: SVF.
- Building %, Ceiling %, Grass %, Plant %, Sky %, Total Builtup %, Total Vegetation %, Tree %, Wall %: Sky openness represented by SVF; scene percentages are separate secondary constructs, not equivalent exposures

9. Sound
- band proportion 20-200 Hz, mechanisms proportion, spectral centroid (Hz), vehicle proportion, wind proportion: Included as an additional sound main effect in Eq2 following the PCA-informed sound screen.
- band proportion 200-2000 Hz, band proportion 2000-20000 Hz, dominant frequency (Hz), octave proportion 1000 Hz, octave proportion 125 Hz, octave proportion 2000 Hz, octave proportion 250 Hz, octave proportion 31.25 Hz, octave proportion 4000 Hz, octave proportion 500 Hz, octave proportion 62.5 Hz, octave proportion 8000 Hz, spectral rolloff 85% (Hz): Not in the PCA-informed Eq2 sound set; spectral information is distinct from sound level.
- category, mechanisms score, rural natural proportion, rural natural score, unidentify proportion, urban manmade negative proportion, urban manmade negative score, urban manmade neutral proportion, urban manmade neutral score, urban manmade positive proportion, urban manmade positive score, vehicle score, wind score: Not in the PCA-informed Eq2 sound set; sound type is distinct from sound level.

10. Solar
- activeness, fitness: Not a continuous numeric exposure; no valid continuous-effect parameterization is declared. The original field remains in the input.
- BVF, TVF: Annular building/tree view fractions; sky openness representative is SVF; not co-entered in core Step-2 screen
- Clearness Index: Derived clearness index (kt = Gh/G0); measured solar exposure is ucm_SUN_Gh [ClearCI p.3 eq.2]
- Clearness Index G0: Calculated extraterrestrial irradiance G0 (date/time/location only), not a measured or personal exposure; denominator of Clearness_Index (kt = Gh/G0) [ClearCI p.3 eq.2]; measured solar exposure is ucm_SUN_Gh.
- Clearness Index Infer: Derived categorical clearness inference from Clearness_Index; not a continuous numeric exposure; no valid continuous-effect parameterization is declared. The original field remains in the input. Measured solar exposure is ucm_SUN_Gh

11. Empatica
- acc x, acc y, acc z, met, vector magnitude: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.
- eda phasic, eda tonic, heart rate, hrv fd hf, hrv fd hfnu, hrv fd lf, hrv fd lf hf ratio, hrv fd lfnu, hrv fd total power, hrv fd vlf, hrv td rmssd, hrv td sdnn, skin temperature: Used as a physiological or ocular response, not an environmental predictor.
- eda scl usiemens: Signal quality, missingness verification, or HRV window integrity metric.

12. Eye-tracker
- blink duration s, distance from center, eyelid aperture avg, fixation duration s, fixation rate, gaze centrality, gaze dispersion, gaze velocity, pupil change rate, pupil diameter avg, saccade amplitude, saccade duration s, saccade peak velocity: Used as a physiological or ocular response, not an environmental predictor.
- stress composite: Signal quality, missingness verification, or HRV window integrity metric.

13. Questionnaire
- current comfort 1 7, environmental comfort mean 1 5, pleasantness 1 5, prs11 being away mean 1 5, prs11 coherence mean 1 5, prs11 fascination mean 1 5, prs11 scope mean 1 5, prs11 total mean 1 5, space familiarity 1 5, stai6 mean 1 5, thermal comfort 1 5: Signal quality, missingness verification, or HRV window integrity metric.

14. Metadata / QC
- age, bike frequency, bmi category who, bmi kg m2, clothing accessories, clothing footwear, clothing height raw, clothing lower body, clothing skin tone, clothing upper body, clothing weight raw, commute modes, daily walking, gender, has children, height m, nationality, own bike for experiment, participant id, regular physical activity, residence type, residential area, response index, site familiarity 1 10, vision correction, weight kg, years in brussels: Baseline participant demographic characteristics or clothing insulation metadata.
- Date, Date (Atmo/LYS), Date (eye), Date (UCM), Datetime, GPS alt, GPS hdg, GPS HDOP, GPS lat, GPS lon, GPS speed valid sample count, GVI match dist m, MAG hdg, ParticipantID, pct complete, PhaseID: Spatiotemporal tracking, GPS positioning, or phase grouping metadata.
- end time s, segment: YAMNet/spectral processing provenance, completeness, timing, or frequency-coverage QC; not an exposure predictor.
- HRV 5min available, HRV 5min max gap (s), HRV 5min removed pct, HRV 5min segment end, HRV 5min segment start, HRV 5min valid rri count, HRV 5min valid span (s): Signal quality, missingness verification, or HRV window integrity metric.
- SVF Frames In Window, SVF Status: SVF calibration, segmentation, run-integrity, telemetry, or manual-validation QC; not an exposure predictor.

STEP 2
Domain Grouping (16)

Thermal	Index	Acoustic	Radiation	Pollution	Light	Greenery
AIR T mid	HUMIDEX	SND dBA	MRT	pm1 avg	lux avg	GVI 50m
AIR RH	PET		SUN Gh	pm2.5 avg	kelvin avg	
WIND AWS	UTCI		SVF	pm10 avg		

STEP 3
Equations + design VIF
(7 equations)

Eq 1: UTCI + sound + global solar radiation + SVF + PM2.5 + lux + colour temperature + GVI + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 2.62
Eq 2: UTCI + sound + global solar radiation + SVF + PM2.5 + lux + colour temperature + GVI + vehicle_proportion + freq_proportion_20_to_200Hz + wind_proportion + spectral_centroid_hz + mechanisms_proportion + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 2.63
Eq 3: air temperature (mid) + sound + humidity + apparent wind + mean radiant temperature + global solar radiation + SVF + PM2.5 + lux + colour temperature + GVI + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (air temperature (mid) x sound) + (sound x humidity) + (sound x apparent wind) + (sound x mean radiant temperature)	max VIF = 2.91
Eq 4: HUMIDEX + sound + global solar radiation + SVF + PM2.5 + lux + colour temperature + GVI + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (HUMIDEX x sound)	max VIF = 1.51
Eq 5: PET + sound + global solar radiation + SVF + PM2.5 + lux + colour temperature + GVI + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (PET x sound)	max VIF = 2.41
Eq 6: UTCI + sound + global solar radiation + SVF + PM1 + lux + colour temperature + GVI + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 2.62
Eq 7: UTCI + sound + global solar radiation + SVF + PM10 + lux + colour temperature + GVI + Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases + (UTCI x sound)	max VIF = 2.62